

Lagrangian Mechanics -- First Pass

PY 501 — Classical Mechanics

The Lagrangian

$L = T - V$, where T is kinetic energy and V is potential energy in generalized coordinates.

Euler–Lagrange equation

$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$, derived from requiring the action to be stationary.

Worked example: simple pendulum

$L = \frac{1}{2} m l^2 \dot{\theta}^2 + mgl \cos\theta$, giving $\ddot{\theta} + (g/l) \sin\theta = 0$.